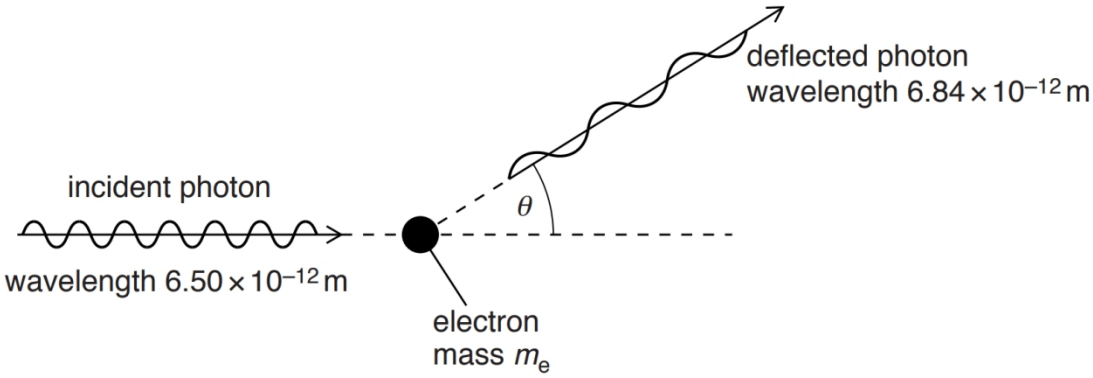
22.2.1 Exercises - Photoelectric effect 1

1. A photon of wavelength 6.50×10^{-12} m is incident on an isolated stationary electron, as illustrated in Fig.



The photon is deflected elastically by the electron of mass m_e. The wavelength of the deflected photon is 6.84×10^{-12} m.

a. Calculate, for the incident photon,

i. its momentum,

momentum = ... Ns [2]

ii. its energy.

energy = ... J [2]

b. The angle θ through which the photon is deflected is given by the expression

$$\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

where Δλ is the change in wavelength of the photon, h is the Planck constant and c is the speed of light in free space.

i. Calculate the angle θ.

θ = ... ° [2]

ii. Use energy considerations to suggest why Δλ must always be positive.

.....

.....

.....[3]

2. For a particular metal surface, it is observed that there is a minimum frequency of light below which photoelectric emission does not occur. This observation provides evidence for a particulate nature of electromagnetic radiation.

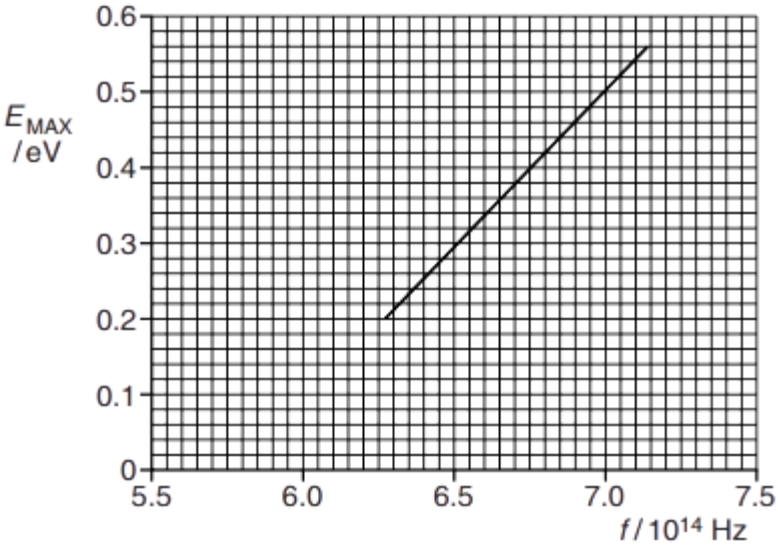
a. State three further observations from photoelectric emission that provide evidence for a particulate nature of electromagnetic radiation.

1.....

2.....

3.....[3]

b. Some data for the variation with frequency f of the maximum kinetic energy EMAX of electrons emitted from a metal surface are shown in Fig.



i. Explain why emitted electrons may have kinetic energy less than the maximum at any particular frequency.

.....

.....[2]

ii. Use Fig. to determine

a) the threshold frequency,

threshold frequency = ... Hz [1]

b) the work function energy, in eV, of the metal surface.

work function energy = ... eV [3]

3.

a. State what is meant by the work function energy of a metal.

.....

.....[2]

b. Ultraviolet radiation of frequency 1.36×10^{15} Hz is incident, in a vacuum, on a metal surface. The power of the radiation incident on the surface is 8.36mW. Photoelectrons are emitted with a maximum kinetic energy of 3.09×10^{-19} J.

i. Determine the number of photons incident on the surface per unit time.

number per unit time = ... s^{-1} [2]

ii. Calculate the work function energy Φ of the metal.

Φ = ... J [2]

c. The frequency of the radiation incident on the surface in (b) is increased while the power remains constant.

State and explain the effect of this change on:

i. the maximum kinetic energy of the photoelectrons

.....

.....[2]

ii. the rate of emission of photoelectrons.

.....

.....[2]

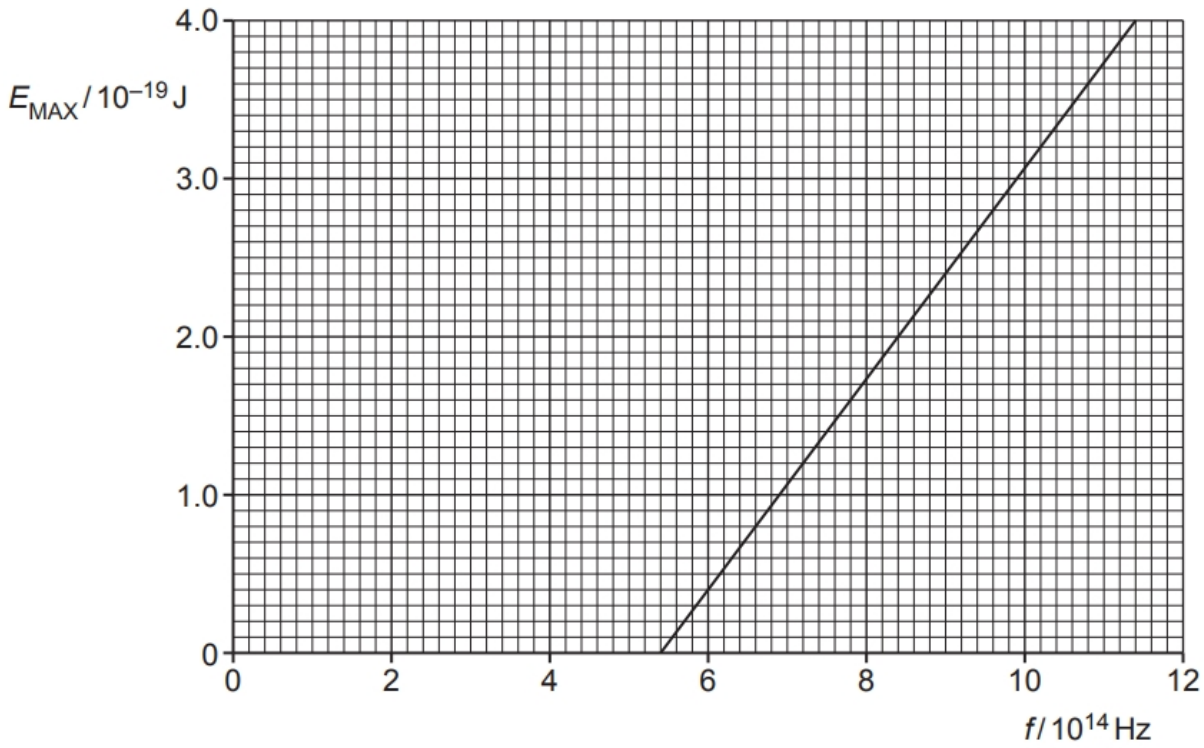
4.

a. State what is meant by a photon.

.....

.....[2]

b. Electromagnetic radiation of a varying frequency f and constant intensity I is used to illuminate a metal surface. At certain frequencies, electrons are emitted from the surface of the metal. The variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons is shown in Fig.



i. State the name of this phenomenon.

.....[1]

ii. Describe three conclusions that can be drawn from the graph in Fig. The conclusions may be qualitative or quantitative.

1.....

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2.....

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3.....

.....[3]

c. The experiment in (b) is repeated twice, each time making one change.

State, with a reason, how the graph obtained would compare with Fig. when:

i. a different metal is used, but keeping the intensity I of the radiation the same

.....

.....[2]

ii. the same metal is used, but with electromagnetic radiation of intensity 2I.

.....

.....[2]